9g. Lazy Broadcasting and Loop Fusion

Martin Alfaro
PhD in Economics

INTRODUCTION

This section continues the analysis of lazy operations as a means of reducing memory allocations. The focus now shifts to broadcasting operations, which strike a balance between code readability and performance.

A key feature of broadcasting in Julia is its default eager behavior. This means that broadcast operations compute and materialize outputs immediately upon execution. As a consequence, broadcasting objects like vectors inevitably leads to memory allocation. Such behavior becomes especially relevant in scenarios with intermediate broadcast operations, whose allocations are potentially avoidable.

To address this performance cost, we'll present two strategies. The first one highlights the notion of **loop fusion**. This enables the combination of multiple broadcasting operations into a single more efficient one. By executing the combined operations in one pass over the data, loop fusion significantly minimizes memory allocations, although not necessarily completely eliminating them.

After this, we'll explore the `LazyArrays` package, which evaluates broadcasting operations **lazily**. The main advantage of this approach is for reductions requiring intermediate calculations. In those cases, a lazy technique can completely bypass memory allocations by avoiding the materialization of temporary arrays.

HOW DOES BROADCASTING WORK INTERNALLY?

Broadcasting operations are essentially syntactic sugar for for-loops, sparing developers from writing them explicitly. This makes it possible to write code that's concise and expressive, without sacrificing performance.

Despite this, you'll often notice performance differences in practice. These discrepancies stem primarily from compiler optimizations, rather than inherent differences between broadcasting and for-loops. The reason is that an operation supporting a broadcast form is automatically revealing additional information about its underlying structure. Consequently, the compiler is allowed to apply more aggressive optimizations. In contrast, for-loops are conceived as a general-purpose construct, precluding the compiler from automatically making such assumptions.

It's worth noting, though, that carefully optimized for-loops always match or exceed the performance of broadcasting. The following code snippets demonstrate this point. The first tab outlines the operation being performed, while the second tab provides a rough internal translation of broadcasting into a for-loop.

The third tab demonstrates the equivalence more directly, using a for-loop that resembles the broadcast code more closely. Its implementation adds the `@inbounds` macro, which broadcasting always applies automatically. The definition of `@inbounds` will be studied in a later section. Its inclusion here is only to illustrate the internal implementation of broadcasting.

BROADCASTING

```
x      = rand(100)

foo(x) = 2 .* x
```

```
julia> @btime foo($x)
  134.005 ns (2 allocations: 928 bytes)
```

ROUGH INTERNAL IMPLEMENTATION

```
x      = rand(100)

function foo(x)
    output = similar(x)

    for i in eachindex(x)
        output[i] = 2 * x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  103.603 ns (2 allocations: 928 bytes)
```

ACTUAL INTERNAL IMPLEMENTATION

```
x      = rand(100)

function foo(x)
    output = similar(x)

    @inbounds @simd for i in eachindex(x)
        output[i] = 2 * x[i]
    end

    return output
end
```

```
julia> @btime foo($x)
  70.333 ns (2 allocations: 928 bytes)
```

> ⚠️ **Warning!** - About `@inbounds`
>
> In the example provided, `@inbounds` **was solely added to demonstrate the internal implementation of broadcasting, not as a general recommended practice**. In fact, `@inbounds` **can cause serious issues** when used incorrectly.
>
> To understand the role of this macro, recall that Julia performs bounds checks in for-loops by default. This means Julia verifies the existence of elements accessed during every iteration. Bounds checking prevents out-of-range access, so that a vector `x` with 3 elements isn't accessed at index `x[4]`. Instead, placing `@inbounds` at the beginning of a for-loop instructs Julia to disable these checks.
>
> Removing bounds checking can improve performance, but it comes at the cost of safety: an out-of-range access may result in program termination or trigger more severe issues. Out-of-bounds access can't occur when operations support broadcasting, since the broadcast mechanism guarantees a valid index range. Consequently, Julia safely omits these checks by automatically applying `@inbounds`.

REMARK (OPTIONAL) Other Optimization Differences Between Broadcasting and For-Loops

CONSEQUENCES OF HOW BROADCASTING IS INTERNALLY COMPUTED

Once we understand how broadcasting is computed internally, we can also anticipate its consequences for memory allocations. First, broadcasting involves the creation of a collection like `output` to store the result. Therefore, the operation will necessarily allocate when broadcasting vectors, since `output` will inherit the type of its inputs.

Importantly, **memory allocations in broadcasting arise even when the result isn't explicitly stored**. For example, evaluating `sum(2 .* x)` involves storing internally the intermediate output `2 .* x`.

Second, Julia's broadcasting is eager by default. Continuing with the same example, this means that `2 .* x` in `sum(2 .* x)` is computed immediately. As we'll see, adopting a lazy strategy can be advantageous in such cases. By deferring the computation of `2 .* x` until `sum(2 .* x)` is executed, we let Julia know that the broadcast operation will be used as part of a reduction. Thus, the compiler can optimize the internal computation, by generating a for-loop that reduces a transformed version of `x`. This avoids materializing the intermediate result altogether, thereby eliminating the temporary allocation for `2 .* x`.

LOOP FUSION

When working with long broadcast operations, splitting them into smaller intermediate steps can significantly improve readability and reduce the likelihood of errors. However, this approach comes at the cost of separately allocating each broadcasting operation.

To mitigate unnecessary memory allocations, it's essential to preserve **loop fusion**: a compiler optimization that merges multiple element-wise operations into a single loop over the data. Thus, with loop fusion enabled, the compiler can perform all operations in a single pass over the data. This not only eliminates the creation of multiple intermediate vectors, but also provides the compiler with a holistic view of the operations, thus unlocking further optimizations.

Loop fusion is applied automatically when all operations are expressed as a single broadcast operation. Yet, as indicated before, writing a single lengthy monolithic expression is inconvenient for complex computations. To overcome this limitation, we can rely on the lazy design of function definitions. Below, we show how this approach can break down an operation into partial calculations, while still preserving loop fusion.

OPERATION SPLIT

```
x        = rand(100)

function foo(x)
    a      = x .* 2
    b      = x .* 3

    output = a .+ b
end
```

```
julia> @btime foo($x)
  293.200 ns (6 allocations: 2.719 KiB)
```

LOOP FUSION

```
x        = rand(100)

foo(x)   = x .* 2 .+ x .* 3      # or @. x * 2 + x * 3
```

```
julia> @btime foo($x)
  114.974 ns (2 allocations: 928 bytes)
```

OPERATION SPLIT + LOOP FUSION

```
x        = rand(100)

term1(a) = a * 2
term2(a) = a * 3

foo(a)   = term1(a) + term2(a)
```

```
julia> @btime foo.($x)
  127.305 ns (2 allocations: 928 bytes)
```

Even with a single simple operation, certain coding patterns can inadvertently break loop fusion. When this occurs, Julia will internally fall back to evaluating sub-expressions in separate for-loops, which are eventually combined to present the final output. In the following, we present two of these cases. The key practical takeaway is that you should broadcast via the macro `@.` when possible, rather than manually adding dots to each operator and function within an expression.

MIXING BROADCASTING WITH VECTOR OPERATIONS BREAKS LOOP FUSION

To understand the issue, there are two facts to consider. The first one is that some vector operations produce an equivalent result to their broadcast counterparts. For instance, adding two vectors with `+` yields the same result as summing them element-wise with `.+`.

BROADCASTING

```
x        = [1, 2, 3]
y        = [4, 5, 6]

foo(x,y) = x .+ y
```

```
julia> foo(x,y)
3-element Vector{Int64}:
 5
 7
 9
```

VECTOR OPERATION

```
x        = [1, 2, 3]
y        = [4, 5, 6]

foo(x,y) = x + y
```

```
julia> foo(x,y)
3-element Vector{Int64}:
 5
 7
 9
```

Another example is with product operations.

BROADCASTING

```
x        = [1, 2, 3]
β        = 2

foo(x,β) = x .* β
```

```
julia> foo(x,β)
3-element Vector{Int64}:
 2
 4
 6
```

VECTOR OPERATION

```
x        = [1, 2, 3]
β        = 2

foo(x,β) = x * β
```

```
julia> foo(x,β)
3-element Vector{Int64}:
 2
 4
 6
```

The second fact to consider is that vector operations don't participate in loop fusion. Thus, even if all these operations were combined into a single expression, Julia will evaluate each sub-expression separately, allocating temporary vectors at every step.

VECTOR OPERATION

```
x = rand(100)

foo(x)  = x * 2 + x * 3
```

```
julia> @btime foo($x)
  394.873 ns (6 allocations: 2.719 KiB)
```

INTERNAL IMPLEMENTATION

```
x = rand(100)

function foo(x)
    term1  = x * 2
    term2  = x * 3

    output = term1 + term2
end
```

```
julia> @btime foo($x)
  312.227 ns (6 allocations: 2.719 KiB)
```

Putting both facts together, mixing broadcasting with vector operations in the same expression may yield the correct result, but only achieve loop fusion partially. This means Julia will only fuse contiguous broadcast segments, internally partitioning the computation into separate for-loops when a vector operation is encountered. Each of these for-loops will produce a temporary intermediate vector.

PARTIAL FUSION

```
x      = rand(100)

foo(x) = x * 2 .+ x .* 3
```

```
julia> @btime foo($x)
  244.920 ns (4 allocations: 1.812 KiB)
```

INTERNAL IMPLEMENTATION

```
x      = rand(100)

function foo(x)
    term1  = x * 2

    output = term1 .+ x .*3
end
```

```
julia> @btime foo($x)
  230.167 ns (4 allocations: 1.812 KiB)
```

This behavior leads to a clear and actionable guideline: for full loop fusion, every operator and function in the expression must be explicitly broadcast. Yet manually adding dots throughout a large expression is both tedious and error-prone: one missing dot is enough to reintroduce unnecessary allocations. There are two coding practices that mitigate this risk.

One option is to broadcast via the macro `@.`, as shown below in the tab "Equivalent 1". By design, this adds a dot to *all* operators and functions within an expression, guaranteeing loop fusion. An alternative is to express the operation through a *scalar* function, which we then broadcast. This is presented below in the tab "Equivalent 2".

LOOP FUSION

```
x      = rand(100)

foo(x) = x .* 2 .+ x .* 3
```

```
julia> @btime foo($x)
  117.949 ns (2 allocations: 928 bytes)
```

EQUIVALENT 1

```
x      = rand(100)

foo(x) = @. x * 2 + x * 3
```

```
julia> @btime foo($x)
  122.426 ns (2 allocations: 928 bytes)
```

EQUIVALENT 2

```
x      = rand(100)

foo(a) = a * 2 + a * 3
```

```
julia> @btime foo.($x)
  124.282 ns (2 allocations: 928 bytes)
```

LAZY BROADCASTING

Broadcasting results in memory allocations since the technique is eager by default. This property means that its output becomes readily available when a broadcast expression is evaluated. Considering this, another way to achieve loop fusion is by evaluating all broadcast sub-expressions lazily, until the final output is computed. The approach is similar in spirit to the use of helper functions to accomplish the same goal, but without cluttering the namespace with new functions.

The functionality is provided by the macro `@~` from the `LazyArrays` package. By prepending this macro to the broadcasting operation, its computation is deferred until its output is required.

EAGER BROADCASTING

```
x         = rand(100)

function foo(x)
    term1  = x .* 2
    term2  = x .* 3

    output = term1 .+ term2
end
```

```
julia> @btime foo($x)
  394.852 ns (6 allocations: 2.719 KiB)
```

LAZY BROADCASTING @~

```
x         = rand(100)

function foo(x)
    term1  = @~ x .* 2
    term2  = @~ x .* 3

    output = term1 .+ term2
end
```

```
julia> @btime foo($x)
  120.214 ns (2 allocations: 928 bytes)
```

SPLITTING OPERATIONS WITH FUNCTIONS

```
x         = rand(100)

term1(a)  = a * 2
term2(a)  = a * 3

foo(a)    = term1(a) + term2(a)
```

```
julia> @btime foo($x)
  134.263 ns (2 allocations: 928 bytes)
```

Beyond this resemblance with an approach based on functions, lazy broadcasting additionally addresses certain scenarios that functions can't handle. To understand this, note that the operations explored thus far resulted in a *vector* output. In those cases, memory allocations could at best be reduced to a single unavoidable allocation, necessary for storing the final output.

When the final output is instead given by a scalar, as occurs with reductions, lazy broadcasting is capable of entirely eliminating memory allocations. The reason is that lazy broadcasting fuses with reduction operations, letting the compiler apply a **non-allocating procedure**. This is illustrated in the example below.

EAGER BROADCASTING

```
# eager broadcasting (default)
x      = rand(100)

foo(x) = sum(2 .* x)
```

```
julia> @btime foo($x)
  120.489 ns (2 allocations: 928 bytes)
```

LAZY BROADCASTING

```
using LazyArrays
x      = rand(100)

foo(x) = sum(@~ 2 .* x)
```

```
julia> @btime foo($x)
  25.216 ns (0 allocations: 0 bytes)
```

> **Lazy Broadcasting May Be Faster Than Other Lazy Alternatives**
>
> An additional advantage of `@~` is that it implements extra optimizations. This explains why `@~` tends to be faster than alternatives like a lazy map, even though neither allocates memory. This performance benefit can be noticed in the following comparison.

LAZY BROADCASTING @~

```
x        = rand(100)

term1(a) = a * 2
term2(a) = a * 3
temp(a)  = term1(a) + term2(a)

foo(x)   = sum(@~ temp.(x))
```

```
julia> @btime foo($x)
  39.288 ns (0 allocations: 0 bytes)
```

LAZY MAP

```
x        = rand(100)

term1(a) = a * 2
term2(a) = a * 3
temp(a)  = term1(a) + term2(a)

foo(x)   = sum(Iterators.map(temp, x))
```

```
julia> @btime foo($x)
  88.509 ns (0 allocations: 0 bytes)
```